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Correction of ^{13}C Mass Isotopomer Distributions for Natural Stable Isotope Abundance

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Metabolism of singly or multiply ^{13}C-labeled substrates leads to the production of molecules that contain ^{13}C atoms at various positions. Molecules differing only in the number of isotopic atoms incorporated are referred to as mass isotopomers. The distribution of mass isotopomers of many molecules can be measured by gas chromatography/mass spectrometry after chemical derivatization. Quantification of metabolite mass isotopomer abundance resulting from biological processes necessitates correction of the measured mass isotopomer distribution of the derivatized metabolite for contributions due to naturally occurring isotopes of its elements. This correction must take into account differences in the relative natural abundance distribution of each mass isotopomer (skewing). An IBM-compatible computer program was developed which (i) calculates the natural abundance mass isotopomer distribution of unlabeled and labeled standards given the molecular formula of the derivatized molecule or fragment ion, and (ii) calculates the natural abundance mass isotopomer distribution of the singly and multiply labeled molecule or fragment via non-linear fitting to the measured mass isotopomer distribution of the unlabeled molecule or fragment. The output of this program is used to correct measured mass isotopomer distributions for contributions from natural isotope abundances and to verify measured values for theoretical consistency. Differences between predicted and measured unlabeled and ^{13}C-labeled isotopomer distributions for hydroxamate di-t-butyldimethylsilyl (di-TBDMS) derivatized pyruvate were measured. The program was applied to the mass isotopomer distribution of glucose labeled from [U-$^{13}C_3$]glycerol and of fatty acids labeled from [U-$^{13}C_6$]glucose and either [2-$^{13}C_2$] acetate or [U-$^{13}C_2$]acetate. In some of these cases, the measured mass isotopomer distributions corrected by the program were different from those corrected by the classical technique. Implications of these differences including those on the calculation of glucose production due to gluconeogenesis in isolated perfused rat liver are discussed.

KEYWORDS: isotopomers; natural enrichment; carbon-13; mass spectrometry; gluconeogenesis

INTRODUCTION

Mass isotopomers or isotopologs of a molecule differ only in the number of atoms of stable isotopes incorporated. Mass isotopomer distribution (MID) analysis is a recently developed technique[1] which has been applied extensively to investigations of lipogenesis,[2–7] gluconeogenesis[8–10] and the regulation of the citric acid cycle.[11] Incorporation of labeled substrates such as [1-^{13}C]acetate or ^2H_2O into fatty acids,[2–7,12,13] [U-^{13}C] acetate or [U-$^{13}C_3$]glycerol into glucose[8–10] or [U-$^{13}C_5$]glutamate into citric acid cycle intermediates[11] results in a significant production of heavy mass isotopomers of metabolic intermediates and products. The MID quantitates the composition of a mass isotopomer mixture. Often the MID of a metabolite can be assayed by gas chromatography/mass spectrometry (GC/MS), the resulting spectrum being a measure of the isotopomer profile of a large population of molecules. The measured MID is dependent on mass isotopomer (i) production via biological processes and (ii) elemental composition. Factor (i) determines the extent of mass isotopomer enrichment in the MID. Factor (ii) includes the derivatized metabolite's molecular formula and the natural abundance isotope distribution for each of its elements. For biological experiments using stable isotope-labeled substrates, the investigator is interested in quantifying factor (i) by accounting for factor (ii).

The largest abundance peak in the GC mass spectrum of an unlabeled (natural abundance) ion determines the ion's base mass. Satellite peaks are present usually at higher masses with some exceptions (e.g. ions with boron). These peaks result from the presence of naturally occurring stable isotopes for each element of the ion.[14] Relative areas of the base and satellite peaks in the mass spectrum can be theoretically predicted from the ion's molecular formula using the stable isotopic distribution of component elements and statistical theory. The elemental distribution for carbon theoretically follows a binomial distribution because there is

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only one stable isotope (^{13}C) of significant abundance (1.11%)[15] present naturally. Considering a large population of molecules each containing n carbon atoms, the distribution of molecules containing zero to n atoms of ^{13}C can be predicted by a binomial distribution based on the natural abundance isotope distribution of carbon. In general, the distribution of molecules containing zero to n atoms of isotopes of an element with several naturally occurring heavy isotopes is predicted by a multinomial distribution. The parameters of elemental binomial and multinomial distributions for a molecule are (i) the number of atoms of each element and (ii) the natural isotopic distribution of each element. Discrete convolutions of elemental frequency distributions result in the predicted MID of the entire fragment. The GC mass spectrum of a population of elementally identical molecules, highly enriched with heavy isotope at one or more atom positions (isotope labeled), is dependent on the extent of this enrichment. Mass spectra of molecules intentionally labeled with heavy isotopes at certain atom positions will possess a characteristic MID that results from heavy isotope incorporation at unlabeled atom sites due to natural isotopic abundance.

Compared with a less labeled isotopomer, an increased occurrence of ^{13}C at one or more carbon positions decreases the relative frequency of higher mass isotopomers in the pure isotopomer molecule's MID. These higher mass isotopomers are associated with natural isotope abundance. Thus, a decrease in the number of carbon atoms associated with ^{13}C natural abundance, i.e. an increase in ^{13}C labeling, decreases the total relative abundance of satellite peaks. Because these MIDs are theoretically based on multinomial distributions, changes in the natural abundance MID with labeling are non-linear.[16] Rosenblatt *et al.*[16] described this phenomenon as a 'skew' in the natural abundance MID of mass isotopomers. For elements having low natural abundance of heavy isotopes, such as hydrogen or oxygen, the difference between natural abundance MIDs of unlabeled and labeled fragments is negligible and consideration of natural abundance MID skew is unnecessary. For elements with non-negligible abundances of naturally occurring stable isotopes, such as carbon, the natural abundance MIDs of labeled isotopomers differ significantly from that of the unlabeled ion.

Enrichment is a result of biological production of mass isotopomers from ^{13}C-labeled substrates. The mole percent enrichment (MPE)[14,17] is a measure of mass isotopomer abundance above that resulting from incorporation of naturally occurring isotopes. The MPE is the mole percentage of a certain mass isotopomer in a mixture after correction for natural isotope abundance. To compute the MPE, the measured MID is adjusted by eliminating naturally occurring isotope contributions. Ideally, this adjustment is performed using the known, pure natural abundance MID of the enriched isotopomers present in the ion cluster.

The classical correction assumes the natural abundance MIDs of unlabeled and ^{13}C-labeled fragments are equal but shifted in mass.[18] This correction requires the measurement of only the natural abundance MID of the unlabeled fragment or compound. This measured natural abundance MID is assumed to equal that of all labeled mass isotopomers of the ion shifted appropriately to higher masses. This mass shift equals the sum of contributions of all heavy isotope atoms for all elements in the molecule. By neglecting the skew effect, the classical correction overestimates the ^{13}C natural abundance contributions of enriched higher mass isotopomers, thereby underestimating enrichment at these masses. The skew of the natural abundance MID[16] should be considered both when (i) the element whose stable isotope is used as a molecular label has a non-negligible naturally occurring abundance (i.e. ^{13}C) and (ii) there is sizable enrichment of higher mass isotopomers in the biologically labeled sample.

Variations in the measured natural natural abundance spectrum of a compound or fragment arise from variations in both GC/MS instrumentation and natural isotopic content of elements within the compound. Calibration graphs can be constructed from the measured MID of standards of unlabeled and labeled isotopomers of different enrichments. However, such standards are costly and often unavailable. Also, calibration graphs for MID should be run with the biological samples or as internal standards. Lastly, computation of MID calibration graphs must take into account the actual degree of isotope labeling.

To alleviate these problems, we developed a computational method that incorporates information from the measured MID of an unlabeled standard to calculate precisely theoretical MIDs of labeled isotopomers, thereby minimizing the number of analytical measurements and obviating the need for multiple labeled standards. This method is incorporated into a user-friendly computer program which is available from the authors. The program calculates the natural abundance MID of unlabeled and labeled compounds or fragments using (i) parameters of natural isotopic abundance obtained from the measured MID of an unlabeled standard and (ii) the fragment's molecular formula. The parameters are obtained using optimization techniques and an objective function that incorporates the measured MID of the unlabeled standard. The program calculates the MID of unlabeled and labeled fragments using reported elemental isotopic abundances[15] and those resulting from parameter optimization.

EXPERIMENTAL

Model development

The elements of the natural abundance correction matrix, **CM**, are the mole fractions of mass isotopomer abundances for pure unlabeled (mass $M+0$) and labeled (masses $M+1$, $M+2$, ...) standards (see Appendix). The effect of MID skew is significant only for compounds labeled with elements having non-negligible natural stable isotope abundance such as ^{13}C. Therefore, the following development assumes that the stable isotope used for labeling purposes is ^{13}C, although the theory applies to any stable isotope. The element in the ith row and jth column of **CM** can be

written as

$$CM_{i,j} = \begin{cases} MF^{i-1}(j-1); & j \geq i \\ 0 & j < i \end{cases} \quad (1)$$

where $MF^i(j)$, $j = 0, 1, \ldots$ refers to the natural abundance MID of the standard with a base mass of $M + i$. Each line of the **CM** matrix represents the MID of a pure standard of unlabeled and labeled isotopomers. This matrix relates the column vector set of uncorrected ($\mathbf{A}_r$) and corrected ($\mathbf{A}_c$) mass spectrometer areas (or mole fractions) by the following equation:

$$\mathbf{A}_r^T = \mathbf{A}_c^T[\mathbf{CM}] \quad (2)$$

The vector of corrected mass isotopomer abundances, $\mathbf{A}_c$, can be obtained by multiplying both sides of this equation by the inverse of the correction matrix **CM**:

$$\mathbf{A}_c^T = \mathbf{A}_r^T[\mathbf{CM}]^{-1} \quad (3)$$

This procedure performs the natural abundance correction of measured GC/MS abundances contained in $\mathbf{A}_r$, resulting in $\mathbf{A}_c$. The mole fraction of each mass isotopomer after correction (equal to molar enrichment) is obtained by normalizing the abundance value for each element of $\mathbf{A}_c$ to the sum of abundances of all elements in $\mathbf{A}_c$ (see Appendix). For a molecule or fragment whose abundances at masses M, $M + 1$, $M + 2$ and $M + 3$ are monitored, the 4×4 correction matrix **CM** becomes

Row 1	$MF^0(0)$	$MF^0(1)$	$MF^0(2)$	$MF^0(3)$
Row 2	0	$MF^1(0)$	$MF^1(1)$	$MF^1(2)$
Row 3	0	0	$MF^2(0)$	$MF^2(1)$
Row 4	0	0	0	$MF^3(0)$

Matrix 1

where each row of the matrix represents the MID (mole fractions) of a pure ^{13}C-mass isotopomer with a base mass number of $M + i$. $MF^i(j)$ is the natural abundance mole fraction of the jth isotopomer above the base mass of the $M + i$ isotopomer (with nominal mass number $M + i + j$). Construction of this matrix requires measurement of the mass spectra of four standards: an unlabeled (row 1), a singly ^{13}C-labeled (row 2), a doubly ^{13}C-labeled (row 3) and a triply ^{13}C-labeled (row 4).

Ideally, the standards should have ^{13}C abundances at each labeled position of 100 at.% enrichment (APE). This seldom being the case, the same m/z must be monitored for all standards to assess the extent of incomplete labeling. To transform the matrix into the upper diagonal form of matrix 1, natural abundance corrections must be made to the labeled isotopomer distributions.

If the MIDs of the three standards needed to construct this matrix are not available, it is often assumed that the MIDs of labeled mass isotopomers equal that of the unlabeled isotopomer. This assumption is referred to as the classical correction described by Biemann.[18] The correction matrix **CM** corresponding to this assumption is

Row 1	$MF^0(0)$	$MF^0(1)$	$MF^0(2)$	$MF^0(3)$
Row 2	0	$MF^0(0)$	$MF^0(1)$	$MF^0(2)$
Row 3	0	0	$MF^0(0)$	$MF^0(1)$
Row 4	0	0	0	$MF^0(0)$

Matrix 2

In general, the mole fractions values $MF^i(j)$ for a given j are not equal for different values of i. This is because the total number of carbon positions associated with natural abundance is decreased by i, i.e. the number of carbon positions occupied by ^{13}C. This decrease leads to a non-linear decrease in the corresponding abundance mole fractions as the fragment becomes more labeled. Correction for natural abundance using matrix 2 in Eqn (3) overestimates natural abundance at higher fragment masses.

The method presented here predicts the MID of ^{13}C-labeled isotopomers from the measured MID of an unlabeled standard by optimizing parameters of elemental isotope abundance to minimize the differences between the measured and predicted mass spectra of the unlabeled standard. These optimized parameter estimates are used to calculate theoretical MIDs of incrementally ^{13}C-labeled standards which in turn are used to construct a theoretical correction matrix. Complete ^{13}C labeling results in a mass shift and a reduction in the number of carbon atoms associated with natural abundance. The MPE of a given mass isotopomer is calculated from the distribution that results after natural abundance correction.

Computer implementation

Our technique has been implemented in a simple, self-contained computer program that calculates natural abundance correction matrices with and without optimal parameter estimation. A user-supplied, standard-form DOS text file is used to specify (i) the global elemental formula of the compound or fragment and (ii) the experimentally determined MID of the unlabeled standard. The program uses this information to optimize estimates of elemental natural isotope abundance that minimize the differences between the measured and predicted MID of the unlabeled standard. These abundances are used to calculate theoretical MIDs for incrementally labeled mass isotopomers. The resulting computer-generated correction matrix can be applied to corresponding biologically labeled samples for natural abundance correction.

The computer program has three main parts: (i) data entry, (ii) natural abundance MID calculation (NATCALC) and (iii) parameter estimation (GRG2).[19] The flow chart of the calculations and the relationships between these parts are shown in Fig. 1. Model parameters are the natural isotopic abundances of the fragment constituent elements. The program outputs a text file containing two correction matrices for each standard of the data input file. The first correction matrix is calculated using reported values for natural elemental abundance parameters.[15] The second matrix is calculated after parameter adjustment via non-linear fitting. Because elemental isotope abundance is expected to change little, elemental natural abundance parameters are constrained to no more than 5% variability from reported values. A squared Pearson product correlation coefficient between the predicted and measured natural MID of the unlabeled standard is calculated with each

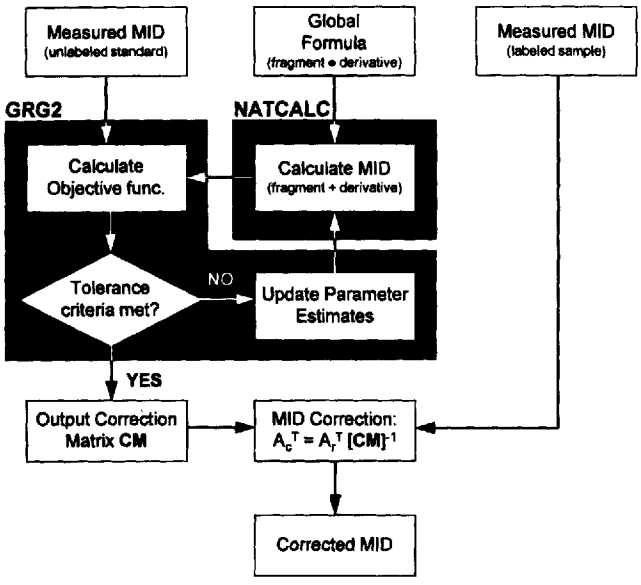


Figure 1. Flow chart showing the relationship between model/parameter estimation routines for generating mass isotopomer distribution correction matrices.

correction matrix. An assessment of the applicability of the computer-generated correction matrix for sample correction can be made from this coefficient. Note that the predicted MID of the unlabeled standard is equal to the first row of each correction matrix. The output text file can be imported into a computer spreadsheet for matrix inversion and subsequent correction of biologically labeled samples for natural abundance. The GRG2 parameter search subroutine[19] implements a reduced gradient method for prameter estimation of linear and non-linear systems. The natural abundance MID calculation subroutine (NATCALC) was written by us and has been validated experimentally and with values from other such programs[16,20] for a wide variety of molecular fragments. NATCALC performs simple discrete convolutions to obtain the natural abundance MID based on statistical theory. Copies of the IBM-compatible software and instructions are available from the authors.

Method validation

To validate this approach, we examined differences between our calculated MIDs and actual GC mass spectra of unlabeled and incrementally labeled pyruvate as its hydroxamate di-TBDMS derivative (molecular formula $C_{11}H_{24}O_3N_1Si_2$).

Applications of method

Lipogenesis. Livers from Sprague–Dawley rats fed a high-sucrose diet were perfused[21] with recirculating albuminated buffer containing 15 mM [U-$^{13}C_6$]glucose and 0.5 mM [2-^{13}C]acetate. Other livers were perfused with 0.5 mM [U-$^{13}C_2$]acetate and 15 mM unlabeled glucose. In both cases, the concentration of [^{13}C]acetate was maintained by constant infusion (1.25 μmol min^{-1}). After 2 h, total fatty acids were extracted from the liver, methylated with diazomethane and assayed by GC electron impact MS.

Gluconeogenesis. Livers from 2-day starved rats were perfused with non-recirculating buffer containing 1 mM lactate and 0.5 mM [U-$^{13}C_3$]glycerol. Glucose was isolated from the effluent perfusate, permethylated[22] and assayed by GC positive-ion chemical ionization (PCI) MS.

The MIDs of derivatized stearate, palmitate and glucose were corrected for natural abundance by the classical[18] and the present methods. Significance at $p < 0.05$ of differences in the corrected MIDs observed between classical and present techniques was assessed using a two-sided, paired Student's *t*-test.

Calculation of fractional gluconeogenesis. Livers from 2-day starved rats were perfused with non-recirculating buffer containing 1 mM [U-^{13}C]pyruvate and 0.1 mM glycerol. Glucose was isolated from the effluent perfusate, permethylated[22] and assayed by GC/PCIMS. For comparison, the MID of glucose was corrected for natural abundance using Biemann's[18] and the present method. Value of fractional gluconeogenesis were calculated from these corrected MIDs using the method of Previs *et al.*[10] Briefly, the mole fractions of glucose mass isotopomers synthesized from labeled triose phosphates, $MF^*_{GLUC}(n)$, can be related to those of triose phosphate intermediates:

$$MF^*_{GLUC}(n) = \sum_{i=1}^{n} MF_{TP}(i)MF_{TP}(n-i); \quad n = 0, 1, \ldots, 6 \quad (4)$$

where $MF_{TP}(i)$ is the mole fraction of the $M + i$ mass isotopomer of triose phosphate. Under the assumption that only one labeled triose phosphate pool of constant enrichment exists and the glucose synthesized from this pool is diluted by unlabeled glucose from other sources, the measured glucose mole fraction MID, $MF_{GLUC}(n)$, can be related to the mole fraction MIDs of newly synthesized and unlabeled glucose by[9,10]

$$MF_{GLUC}(n) = fMF^*_{GLUC}(n) + (1-f)MF^0_{GLUC}(n); \quad n = 0, 1, \ldots, 6 \quad (5)$$

where $MF^0_{GLUC}(n)$ represents the known isotopomer mole fractions of unlabeled glucose and f is the fraction of total glucose production that comes from gluconeogenesis, i.e. via triose phosphates. Note that the measured glucose MID is always corrected for natural abundance using the methods discussed here. Simultaneous solution of Eqns (4) and (5) yields values of $MF_{TP}(i)$ and f compatible with the measured glucose MID.[9,10]

RESULTS AND DISCUSSION

Table 1 compares (i) the correction matrix determined by GC/MS analysis of the hydroxamate di-TBDMS derivatives of unlabeled, [1-^{13}C]- and [2,3-^{13}C]pyruvate and [U-$^{13}C_3$]pyruvate with (ii) the correction

Table 1. Correction matrices for pyruvate hydroxamate di-TBDMS ($C_{11}H_{24}O_3Ni_1Si_2$)

		Mass			
		274	275	276	277
			Mass isotopomer		
Matrices*	Label	$M+0$	$M+1$	$M+2$	$M+3$
A. Matrix of MIDs of GC/MS-measured unlabeled and labeled standards	Unlabeled	0.746	0.169	0.071	0.011
	[1-$^{13}C_1$]	0.006	0.753	0.158	0.070
	[2,3-$^{13}C_2$]	0.000	0.009	0.753	0.156
	[U-$^{13}C_3$]	0.000	0.000	0.020	0.751
B. Matrix corrected for incomplete labeling	Unlabeled	0.746	0.169	0.071	0.011
	[1-$^{13}C_1$]	0.000	0.758	0.159	0.070
	[2,3-$^{13}C_2$]	0.000	0.000	0.760	0.157
	[U-$^{13}C_3$]	0.000	0.000	0.000	0.768
C. Matrix calculated from parameter optimization (CORMAT)	Unlabeled	0.746	0.169	0.071	0.011
	[1-$^{13}C_1$]	0.000	0.754	0.163	0.070
	[2,3-$^{13}C_2$]	0.000	0.000	0.763	0.157
	[U-$^{13}C_3$]	0.000	0.000	0.000	0.771
D. Absolute % differences between elements of matrices B and C	Unlabeled	0.021	0.001	0.005	0.019
	[1-$^{13}C_1$]	0.000	0.373	0.439	0.040
	[2,3-$^{13}C_2$]	0.000	0.000	0.207	0.006
	[U-$^{13}C_3$]	0.000	0.000	0.000	0.276

* Each row of matrix A is the GC/MS-measured MID for the listed unlabeled and labeled standards expressed as mole fractions. Matrix B is the result of correction at each mass of matrix A for incomplete labeling. Matrix C is calculated by our program using only the first line of matrix A (the MID of the unlabeled standard) as input. The absolute percentage differences between the corresponding elements of matrices B and C are the elements of matrix D.

matrix calculated by the present method from the GC/MS analysis of the unlabeled pyruvate derivative ($C_{11}H_{24}O_3N_1Si_2$, $m/z = 274$). Each row of matrix A contains the measured MID of the standards expressed as mole fractions. Note that these standards are not fully labeled, as shown by non-zero abundances at masses below each labeled standard's base mass. Correction of matrix A for this incomplete labeling yields matrix B. Matrix C is the output of our program which uses as input only the first line of matrix A. Matrix D shows the percentage difference between the elements of matrix B and matrix C. Note that the calculation of matrix C requires only the isotopomer distribution of an unlabeled standard. In contrast, generating matrix B requires (i) measurements of the MIDs of three unlabeled and labeled standards and (ii) a subsequent correction for incomplete labeling. There is good agreement between the experimental and estimated values (matrix D).

Figures 2–4 compare the MIDs of selected metabolites corrected for natural abundance using the classical and present techniques. These metabolites, either glucose (Fig. 2) or a fatty acid (Figs 3 and 4), were labeled by perfusion of liver with various ^{13}C substrates. In all cases, significant differences ($p < 0.05$, paired t-test) are observed between the two correction techniques.

Figure 2 shows the MID of glucose highly labeled from [U-$^{13}C_3$]glycerol. The superiority of the present method over the classical method is indicated by the negative value (−3.4%) resulting from the classical correction for the $M + 4$ mass isotopomer of glucose (Fig. 2). This shows overcorrection by the classical method, owing to high enrichment in $M + 3$ isotopomers. Although differences between these MIDs appear small, calculation of parameters in models of gluconeogenesis can be affected. One such parameter, the fraction of glucose production due to gluconeogenesis, f,[9,10] was calculated from the MID of glucose made in livers perfused with [U-^{13}C]pyruvate. The MID of glucose resulting from the correction of Biemann and that described here are shown in Table 2. Also shown in Table 2 is the value of f calculated from these distributions.[10] The value of f after correction by the present

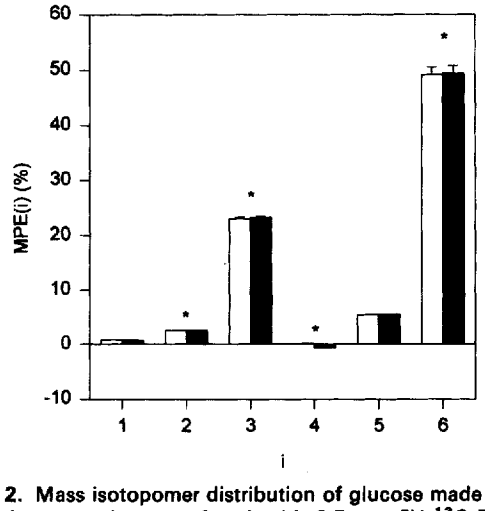


Figure 2. Mass isotopomer distribution of glucose made in livers from 2-day starved rats perfused with 0.5 mM [U-$^{13}C_3$]glycerol and 1.0 mM lactate. The vertical scale represents labeled mass isotopomer mole fraction $MF_e(i)$ and the horizontal scale represents the number of mass units of the labeled isotopomer above that of the unlabeled one (i). Comparison of natural abundance corrections using the present method (open bars) and classical correction[18] (hatched bars). * Statistically significant ($p < 0.05$) using two-sample, paired t-test assuming equal variances. Data are presented as mean MPE ± standard error of three injections.

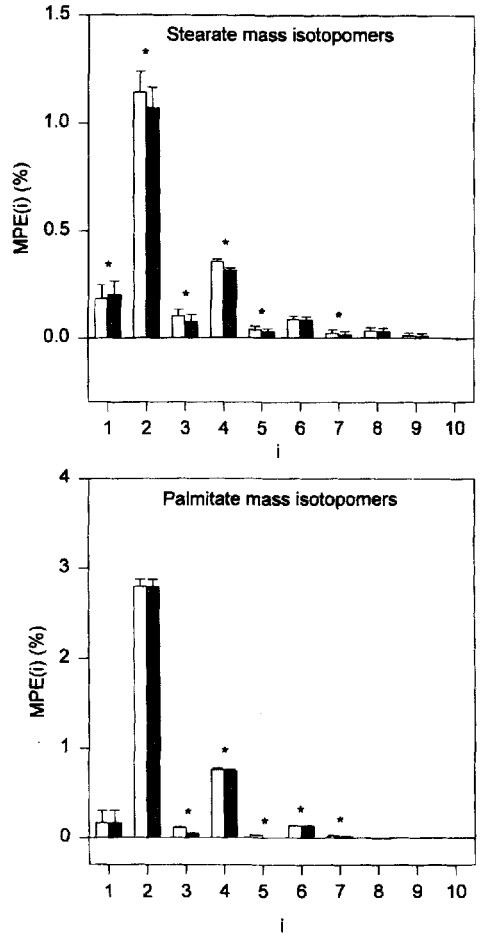


Figure 3. Mass isotopomer ditribution of stearate and palmitate made in livers from high-carbohydrate fed rats perfused with 0.5 mM [U-$^{13}C_2$]acetate and 15 mM glucose. Comparison of natural abundance corrections using present method (open bars) and classical correction[18] (hatched bars). * Statistically significant ($p < 0.05$) using two-sample, paired *t*-test assuming equal variances. Data are presented as mean MPE ± standard error of four injections.

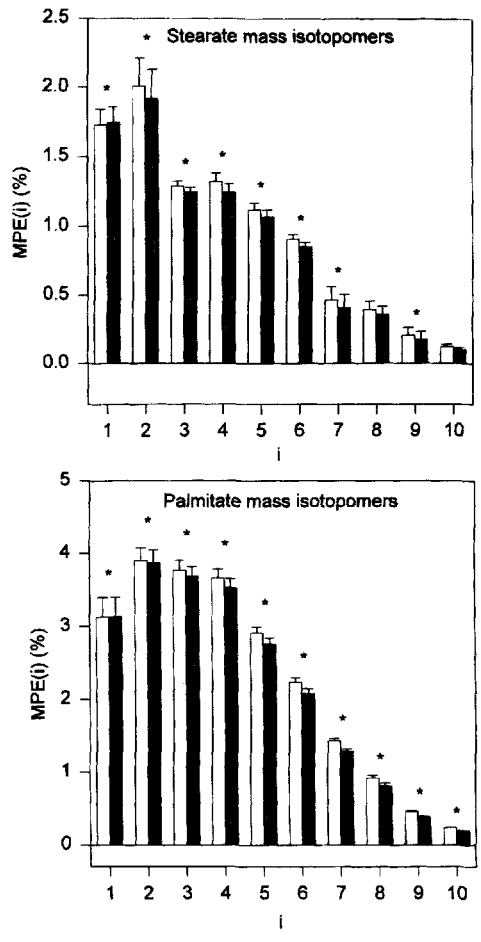


Figure 4. Mass isotopomer distribution of stearate and palmitate made in livers from high-carbohydrate fed rats perfused with 15 mM [U-$^{13}C_6$]glucose and 0.5 mM [2-$^{13}C_2$]acetate. Comparison of natural abundance corrections using present method (open bars) and classical correction[18] (hatched bars). * Statistically significant ($p < 0.05$) using two-sample, paired *t*-test assuming equal variances. Data are presented as mean MPE ± standard error of four injections.

Table 2. Comparison between the MID of glucose labeled from [U-^{13}C]lactate in perfused rat livers after correction for contributions from natural stable isotope abundance using the method of Biemann[18] and that described in the text

Mass isotopomer	Glucose MID	
	Biemann method[18]	Present method
M + 0	53.63 ± 3.88	53.0 ± 3.89
M + 1	7.91 ± 0.61	7.81 ± 0.59
M + 2	15.01 ± 1.13	14.90 ± 1.11
M + 3	15.64 ± 1.27	15.76 ± 1.26
M + 4	3.13 ± 0.39	3.59 ± 0.42
M + 5	3.21 ± 0.37	3.31 ± 0.38
M + 6	1.47 ± 0.17	1.62 ± 0.19
f[a]	63.7 ± 4.5	72.4 ± 4.6

[a] The calculation[10] of the percentage contribution of gluconeogenesis to glucose production (*f*) is described under Experimental.

method (72.4%) is higher than that calculated from the MID corrected by the standard method (64.7%, $p = 0.13$). Presumably, the classical method of Biemann overestimates the contribution from natural abundance in the glucose spectrum and thus underestimates the extent of labeling. Because this labeling is derived from gluconeogenesis, *f* is also underestimated.

Figures 3 and 4 show MIDs of stearate and palmitate isolated from rat livers perfused with either unlabeled glucose + [1,2-$^{13}C_2$]acetate (Fig. 3) or [U-$^{13}C_6$]glucose + [2-^{13}C]acetate (Fig. 4). These protocols were chosen to generate different fatty acid MID patterns. In both cases, the classical correction introduces significant errors in the corrected abundances at many masses (Figs 2 and 3). Presumably, as is the case for glucose, these differences could affect calculations of fractional fatty acid synthesis.[2–7]

Our technique should increase the precision of parameters calculated from various types of MID analysis. In studies of the synthesis of labeled polymers, the fractional *de novo* synthesis is calculated from ratios of mass isotopomer abundances which are sensitive to small variations in the elements of that ratio. The classical correction of MIDs affects the precision estimates

of fractional *de novo* synthesis. Note that in their studies of fatty acid and cholesterol synthesis, Hellerstein and co-workers[2,3] calculated fractional *de novo* synthesis in a manner that does not require a correction matrix but is dependent on GC/MS measurement of the MID of an unlabeled standard. In their method, a theoretical calibration graph of excess mass isotopomer ratio *vs.* precursor enrichment accounts for the effect of natural abundance MID skew. However, any discrepancies between the theoretical and measured MID of the unlabeled standard lead to errors that are propagated throughout the calculations.

Previous methods that have examined the separation of isotopomer spectra into individual components include those of Brauman,[23] Lee and co-workers[24,25] and Berthold *et al.*[20] Brauman[23] described a method for resolving mass spectra of a mixture of molecules in which only a single element (chlorine) possesses a natural isotope abundance ditribution.[22] The general case for a molecule containing several atoms of a multi-isotope element or elements was not developed. Lee and co-workers[24,25] described a method for MID correction that uses theoretically derived correction matrices that account for natural abundance MID skewing. This method was developed only for theoretical six-carbon 'glucose' distributions and involves no parameter fitting or MID measurement of standards.[24,25] Berthold *et al.*[20] presented a method that involves no parameter fitting but can calculate theoretical natural abundance distributions for user-specified compounds.[20] Rosenblatt *et al.*[16] thoroughly analyzed the skewing of natural abundance MIDs. To correct for skewing, they derived explicit expressions that use correction factors based on reported natural abundances of constituent atoms. These correction factors account for the skew when calculating MPE. The output of this calculation is then used to derive correction factors that are applied to measured MIDs. Except for the simplest cases, these expressions are computationally daunting for the average biological investigator. Our technique includes simple, statistically based and user-friendly software for performing natural abundance corrections that account for MID skew. The mathematical details of the correction used in our technique are elaborated in the Appendix. In our method, the measured natural abundance MID of an unlabeled standard is incorporated into the correction via non-linear fitting. Derivation of correction factors unique to the derivatized fragment is unnecessary. Our technique ensures a high level of computational precision and efficiency while eliminating complex algebraic manipulation for explicit solutions. In addition, it aids in the detection of cluster contamination or faulty peak integration by comparison of the measured and the predicted unlabeled standard MIDs.

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APPENDIX

Derivation of correction matrix

This Appendix shows the development of Eqns (1)–(3) and matrix 1 in the text. These equations and the correction matrix **CM** are necessary to perform the correction of measured mass isotopomer distributions (MID) for natural abundance. Enrichment is defined as abundance above natural abundance and is expressed as mole percent enrichment (MPE).

Let us define the following isotopomer abundance peak area distributions:

unlabeled standard:	$A^0(0)$, $A^0(1)$	(A1)
singly labeled standard:	$A^1(0)$, $A^1(1)$	(A2)
labeled sample measured:	$A(0)$, $A(1)$, $A(2)$	(A3)
labeled sample corrected:	$A_c(0)$, $A_c(1)$, $A_c(2)$	(A4)

For simplification of the equations, we assume here that there is no significant natural abundance above the $M + 1$ mass isotopomer in each standard ($A^0(i)$ and $A^1(i) = 0$ for $i > 1$). A general treatment yields the same conclusions. Let us assume we have a sample mixture in which the measured non-zero peak areas $A(0)$, $A(1)$ and $A(2)$ arise only from a mixture of unlabeled and singly labeled isotopomers. The abundance at mass $M + 0$ [$A(0)$] is that of unlabeled isotopomer. The abundance of $M + 1$ [$A(1)$] is the sum of enrichments with singly labeled isotopomer and $M + 1$ natural abundance associated with unlabeled isotopomer. Thus, even when the sample mixture consists of only unlabeled material, abundances at masses higher than $M + 0$ will be non-zero owing to natural abundance contributions associated with the unlabeled isotopomer. Abundance at $M + 2$ [$A(2)$] results from natural abundance associated with enrichment of singly labeled isotopomer only. This is because $A^0(2)$, the natural abundance at mass number $M + 2$ associated with unlabeled isotopomer, is assumed equal to zero. Unlabeled isotopomer natural abundance at $M + 2$ and singly labeled natural abundance at $M + 3$ are assumed to be zero. $A_c(0)$ and $A_c(1)$ are the abundances of unlabeled and singly labeled isotopomers corrected for natural abundance and thus due to enrichment only. Measured abundances can thus be expressed as the sum of contributions from enrichment and natural abundance. In a sample isotopomer mixture, we can quantify the natural abundance contribution by using standards of pure unlabeled and labeled isotopomers.

The mole fraction distribution can be calculated from the abundance distribution of the two standards [(A1) and (A2)]:

$$MF^0(0) = A^0(0)/[A^0(0) + A^0(1)] \quad \text{(A5)}$$

$$MF^0(1) = A^0(1)/[A^0(0) + A^0(1)] \quad \text{(A6)}$$

$$MF^1(0) = A^1(0)/[A^1(0) + A^1(1)] \quad \text{(A7)}$$

$$MF^1(1) = A^1(1)/[A^1(0) + A^1(1)] \quad \text{(A8)}$$

These standard mole fractions relate the measured [(A3)] and corrected [(A4)] sample abundances:

$$A(0) = MF^0(0)A_c(0) \quad \text{(A9)}$$

$$A(1) = MF^0(1)A_c(0) + MF^1(0)A_c(1) \quad \text{(A10)}$$

$$A(2) = MF^1(1)A_c(1) \quad \text{(A11)}$$

These expressions correspond to the following correction matrix:

$$CM = \begin{bmatrix} MF^0(0) & MF^0(1) & 0 \\ 0 & MF^1(0) & MF^1(1) \\ 0 & 0 & MF^2(0) \end{bmatrix} \quad \text{(A12)}$$

Note that the value of $MF^2(0)$, the mole fraction at base mass of a doubly labeled standard, is irrelevant as long as there is no $M + 2$ isotopomer enrichment. Solving Eqn (A9) for $A_c(0)$, we obtain

$$\begin{aligned} A_c(0) &= A(0)[A^0(0) + A^0(1)]/A^0(0) \\ &= A(0)[1 + A^0(1)/A^0(0)] \end{aligned} \quad \text{(A13)}$$

and solving Eqn (A10) for $A_c(1)$ we obtain

$$A_c(1) = \{A(1) - [A^0(1)/A^0(0)]A(0)\}[1 + A^1(1)/A^1(0)] \quad \text{(A14)}$$

The expressions for MPE at each mass can now be written:

$$\text{MPE}(0) = 100\{A_c(0)/[A_c(0) + A_c(1)]\} \quad \text{(A15)}$$

$$\text{MPE}(1) = 100\{A_c(1)/[A_c(0) + A_c(1)]\} \quad \text{(A16)}$$